1.) Explain the structure of HTML file with example. How is HTML different from CSS? Explain three different ways for inserting CSS code into a HTML file.

→ The structure of HTML file is

```
<html>
    <head>
        <title> Title of Page </title>
    </head>
    <body>
        content of the page
    </body>
</html>
```

In the above HTML documents, it has two main parts as

i) head :- the head elements contains title and meta data of a web document

ii) body :- the body element contains the information that you want to display on a web page

Moreover,

- <html> indicates that this web page is written in HTML
- <title> tag contains the web page title
- <head> tag contains information about the web
- <body> tag contains the content of the web page
- </html> marks the end of the web page.

SIT Tutor

	HTML	CSS
Definition:-	HTML is the standard mark-up language for describing the structure of web pages.	CSS is the stylesheet language for describing the presentation and design of web pages including colors, fonts & layouts.
Implementation	It is for content and web page structure	It is mainly for presentation and design
Architecture	It use to have tags that are surrounding the content.	In CSS, selectors are declared using block statement syntaxes.
Dependency	It's syntax and structure formats can't be used in CSS stylesheet.	It is independent of HTML and it can be used with any XML-based markup ~~ta~~ language
Approach	HTML mainly used to develop the basic content of any web page	It is mainly implemented for web designing, page style formats, layouts and other visual effects.
→	It is easy to learn	It can some time get messy and can create complications in codes.
→	It takes some time to load.	It loads fast

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The three different ways for inserting CSS code into an HTML file are

i) Inline - by using the style attribute in HTML elements

- It is used to apply unique style to a single HTML element.
- It uses the style attribute of an HTML element
- This type of CSS can not be reused.

Eg :- <h1 style = "color : red ;" > Red </h1>

ii) Internal - by using a <style> element in <head> section

- It is used to define a style for a single HTML page
- It is defined in the <head> section of an HTML page, within a <style> element.
- It can be reused in a single page only

Eg :-
```
<head>
    <style>
        h1 { color : red ; }
    </style>
<body>
    <h1> Red </h1>
</body>
```

iii) External - by using an external CSS file

- It is used to define the style for many pages.
- By changing one file, we can change the look of entire web.
- For this, we add link in the <head> section as

```
<head>
  <link rel = "stylesheet" href = "styles.css">
</head>
```

2.) Write HTML script for creating a form containing text box for usesnare, password field for password, and checkbox for Education fields. Write a JavaScript function for the validation of the form for all of the fields as required. In addition length of usesnare should at least 4, the password should start with digit and end with $.

THE CODE IS AT :-

github.com/aistha11/BscCSIT/tree/master/sem5/WebTech/modelques/qn 2

Things to learn, from this question

i) Use required attribute as
<input type="text" name="usesname" required>

ii) The regular expression for password

var regx = /[0-9][a-zA-Z0-9]*[a-zA-Z0-9][$]/;

which validates that,
1, anything, $ is valid

4.) Describe client server architecture with its types.

→ Client / Server architecture is a computing model in which the server hosts, delivers and manages most of the resources and services to be consumed by the client.
It has one or more clients connected to a central server over a network or internet connection.
WWW is based on this architecture.

Types of client-server architecture

i) One Tier
- the user interface, marketing logic and data logic are present in the same system
- This kind of service is reasonable
- The data is usually stored in the local system
- Completely unscalable. Only one user can access the system at a given time via the local client.

ii) Two Tier
- the UI is stored at client machine and the database is stored on the server.
- Client and Server have to come in direct incorporation
- This is done for rapid results and to avoid confusion between different clients
- It is considered as affordable architecture
- Multiple users can connect to the server at one
- It is not suitable for security reasons.

iii) Multi-Tier

- N-tier architecture (with N more than 2) is really 3 tier architecture in which the middle tier may or may not be split into new tiers.
- The application tier is broken down into seperates parts
- They make load balancing possible
- Processing can be more evenly distributed among servers.
- Are more easily scalable
- It is more difficult to program.

5.) Write an HTML script to insert audio file in a HTML page? Use the controls and autoplay properties.

→ Code At :-
github.com/aisthall/BscCSIT/tree/master/sem5/WebTech/modelques/qn5

7.) Write a JavaScript code to create an array of elements tcsit, it, bca, bim9. Display the array in body of HTML.

→ Code at :-
github.com/aisthall/BscCSIT/tree/master/sem5/WebTech/modelques/qn7

6.) What is CSS selectors ? Compare id selector with class selector.

→ A CSS selector is the first part of a CSS Rule. It is a pattern of elements and other terms that tell the browser which HTML elements should be selected to have the CSS property values inside the rule applied to them. It select HTML elements according to its id, class, type, attribute etc.

Id Selector	Class Selector
→ The id selector is used to specify a style for a single, unique element.	The class selector is used to specify a style for a group of elements.
→ It uses id attribute of the HTML	It uses the HTML class attribute
→ It is defined with "#"	It is defined with "."

→ Eg :-
 #para {
 color : red ;
 }
	Eg :-
	 .para {
	 color : red ;
	 }

→ It is less used	It is most used on several elements.

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8.) Write a HTML script containing use of media query for changing the background-color of html page to black if the viewport is 600 px wide or more than that otherwise if the viewport is less than 500 pixels, the background-color should be changed ~~d~~ to red.

→ Code at :-
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12) Write HTM script for displaying div with following output. The div should be floating right.

CSIT
1. PMC
2. BMC
BCA
- TC
- PPL

→ Code at

github.com/aistha11/BscCSIT/tree/master/sem5/WebTech/modelques/qn 12

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9.) What is XML namespace? How it is used in XML files.

→ XML namespace is a mechanism to avoid name conflicts by differentiating elements or attributes within an XML document that may have identical names, but different definitions.

XML namespace is used in XML files as

i) Using a Prefix

Name conflicts in XML can ~~be~~ easily be avoided using a name prefix. For eg:-

```
<h: table >
    <h: tr >
        <h : td > Apples </h: td>
        <h : td > Banana </h : td >
    </h : tr >
</h : table >
<f: table >
    <f: name > Coffee </f: name>
    <f: width > 80 </f: width >
    <f: length > 120 </f: length>
</f: table >
```

Here, there will be no conflict because the two <table> elements have different names.

ii) Using xmlns Attribute

When using prefixes in XML, a namespace for the prefix must be defined. The namespace can be defined by an xmlns attribute in the start tag of an element. For eg :-

```
<root>
  <h:table xmlns:h="https://hello.com/world">
  <h:tr>
    <h:td> Apples </h:td>
    <h:td> Bananas </h:td>
  </h:tr>
  </~~table~~ h:table>

  <f:table xmlns:f="https://jonjans.com/shoe/">
    <f:name> Shoe World </f:name>
    <f:contact> 9886319931 </f:contact>
  </f:table>

</root>
```

The purpose of using an URI is to give the namespace a unique name.

Namespace is used as a pointer to a webpage containing namespace information.

10.) Discuss different types of jQuery Selectors with examples.

→ The different types of jQuery Selectors are

i) Element Selector :-
- selects elements based on the element name
- ~~When a~~ For eg :- when a user clicks on a button, all <p> elements will be hidden

```
<script>
    $(document).ready(function (){
        $("button").click(function (){
            $("p").hide();
        });
    });
</script>
```

ii) Id selector
- uses the id attribute of an HTML tag to find the specific element.
- For eg :- when a user clicks on a button, the element with id = "test" will be hidden

```
<script>
    $(document).ready(function (){
        $("button").click(function (){
            $("#test").hide();
        });
    });
</script>
```

iii) Class Selector
- finds elements with a specific class
- For eg:- when user clicks on button, the elements with class = "test" will be hidden

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $(".test").hide();
    });
});
</script>
```

SIT Tutor

11.) Write a PHP script that illustrates handling of cookie.

Code at :-

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3.) Prepare a form that should contain text box, selection list and radio button. Write PHP script to store data from the form into database using database connection and appropriate query.

Code at :-
github.com/aisthall/BscCSIT/tree/master/sem5/WebTech/modelques/qn 3